

MICROPROCESSOR PROGRAMMING**Course Code : 314321**

Programme Name/s : Artificial Intelligence/ Artificial Intelligence and Machine Learning/ Computer Technology/ Computer Engineering/ Computer Software Technology/ Computer Science & Engineering/ Data Sciences/ Computer Hardware & Maintenance/ Computer Science

Programme Code : AI/ AN/ CM/ CO/ CST/ CW/ DS/ HA/ SE

Semester : Fourth

Course Title : MICROPROCESSOR PROGRAMMING

Course Code : 314321

I. RATIONALE

The microprocessor is the most vital component of a computer system and is considered be its' brain and heart. This course will cover the basics of 8086 and its architecture along with instruction set, data types, assembly language programming with effective use of procedure and macro. This course will enable the students to inculcate assembly language programming concepts and methodology to solve problems related with microprocessor-based systems.

II. INDUSTRY / EMPLOYER EXPECTED OUTCOME

This course aims to help the student to attain the following industry expected outcomes through various teaching-learning experiences:

*Develop assembly language programs using 8086.

III. COURSE LEVEL LEARNING OUTCOMES (COS)

Students will be able to achieve & demonstrate the following COs on completion of course based learning

- CO1 - Analyze the functional block diagram of 8086 microprocessor.
- CO2 - Use program development tools and assembler directives.
- CO3 - Use instructions in different addressing modes.
- CO4 - Develop an assembly language program for a given task using assembler.
- CO5 - Use procedures and macros to develop an assembly language program for a given problem.

IV. TEACHING-LEARNING & ASSESSMENT SCHEME

Course Code	Course Title	Abbr	Course Category/s	Learning Scheme					Credits	Assessment Scheme											
				Actual Contact Hrs./Week			SLH	NLH		Paper Duration	Theory				Based on LL & TL				Based on SL		Total Marks
															Practical						
				CL	TL	LL					FA-TH	SA-TH	Total		FA-PR		SA-PR		SLA		
															Max	Min	Max	Min	Max	Min	
314321	MICROPROCESSOR PROGRAMMING	MIC	DSC	3	-	2	1	6	3	3	30	70	100	40	25	10	25@	10	25	10	175

Total IKS Hrs for Sem. : 0 Hrs

Abbreviations: CL- Classroom Learning , TL- Tutorial Learning, LL-Laboratory Learning, SLH-Self Learning Hours, NLH-Notional Learning Hours, FA - Formative Assessment, SA -Summative assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment

Legends: @ Internal Assessment, # External Assessment, *# On Line Examination , @\$ Internal Online Examination
Note :

1. FA-TH represents average of two class tests of 30 marks each conducted during the semester.
2. If candidate is not securing minimum passing marks in FA-PR of any course then the candidate shall be declared as "Detained" in that semester.
3. If candidate is not securing minimum passing marks in SLA of any course then the candidate shall be declared as fail and will have to repeat and resubmit SLA work.
4. Notional Learning hours for the semester are (CL+LL+TL+SL)hrs.* 15 Weeks
5. 1 credit is equivalent to 30 Notional hrs.
6. * Self learning hours shall not be reflected in the Time Table.
7. * Self learning includes micro project / assignment / other activities.

V. THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
1	TLO 1.1 Describe the function of the given pin of 8086. TLO 1.2 Explain function of Bus Interface Unit and Execution Unit in 8086 Microprocessor. TLO 1.3 State functions of the given Register of 8086 Microprocessor. TLO 1.4 Calculate the physical address for the given segmentation of 8086 Microprocessor.	Unit - I 8086-16 Bit Microprocessor 1.1 8086 Microprocessor: Salient features, pin descriptions 1.2 Architecture of 8086: Functional block diagram, register organization 1.3 Concept of pipelining 1.4 Memory segmentation, Physical memory addresses generation	Lecture using chalk-board Presentations Hands-on
2	TLO 2.1 Describe the given steps of program development and execution. TLO 2.2 Write steps to develop a code for the given problem using assembly language. TLO 2.3 Use relevant command of debugger to correct the specified programming error. TLO 2.4 Describe function of the given assembler directives with example.	Unit - II The Art of Assembly Language Programming 2.1 Program development steps: Problem definition, Algorithm, Flowchart, Initialization checklist, Choosing instructions, Converting algorithm into assembly language program 2.2 Assembly Language Programming Tools: • Editor • Assembler • Linker • Debugger 2.3 Assembler directives	Lecture using chalk-board Presentations Hands-on Collaborative learning

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
3	TLO 3.1 Determine the length of the given instruction. TLO 3.2 Describe the given addressing modes with examples. TLO 3.3 Explain the operation performed by the given instruction during its execution. TLO 3.4 Identify the addressing mode of the given instruction.	Unit - III Instruction Set of 8086 Microprocessor 3.1 Machine language instruction format 3.2 Addressing modes 3.3 Instruction set: • Arithmetic instructions • Logical Instructions • Data transfer instructions • Flag manipulation instructions • String operation instructions • Program control transfer or branching instructions • Process control instructions	Lecture using chalk-board Presentations Hands-on Collaborative learning
4	TLO 4.1 Use the given model of assembly language program for the given problem. TLO 4.2 Develop ALP for the given problem. TLO 4.3 Apply relevant control loops in the program for the given problem. TLO 4.4 Use string instruction to manipulate the elements of the given block of data.	Unit - IV Assembly Language Programming 4.1 Models of 8086 assembly language program 4.2 Programming using assembler: • Arithmetic operations on hexadecimal and BCD numbers • Sum of series • Smallest and largest numbers from array • Sorting numbers in ascending and descending order • Check whether given number is odd or even • Check whether given number is positive or negative • Block transfer • String operations - Length, Reverse, Compare, Concatenation, Copy • Count numbers of 'I' and 'O' in 16 bit number	Lecture using chalk-board Presentations Hands-on Collaborative learning
5	TLO 5.1 Apply the relevant 'parameter- passing' method in the given situation. TLO 5.2 Develop an assembly language program using the relevant procedure for the given problem. TLO 5.3 Develop an assembly language program using macros for the given problem. TLO 5.4 Compare procedures and macros on the basis of the given parameter.	Unit - V Procedure and Macro 5.1 Procedure: Defining and calling procedure - PROC, ENDP, FAR and NEAR Directives; CALL and RET instructions; Parameter passing methods, Assembly language programs using procedure 5.2 Macro: Defining macro, MACRO and ENDM Directives, Macro with parameters, Assembly language programs using macro	Lecture using chalk-board Presentations Hands-on Collaborative learning

VI. LABORATORY LEARNING OUTCOME AND ALIGNED PRACTICAL / TUTORIAL EXPERIENCES.

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 1.1 Identify the functions of various blocks in 8086 architecture. LLO 1.2 Identify the use of registers of 8086.	1	* Identification of various blocks in 8086 microprocessor architecture	2	CO1
LLO 2.1 Identify the function of given assembly language tool. LLO 2.2 Use assembler directives in a given situation.	2	* Use assembly language programming (ALP) tools and directives	2	CO2

MICROPROCESSOR PROGRAMMING**Course Code : 314321**

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 3.1 Use different addressing mode instructions in program. LLO 3.2 Write an assembly language program for addition and subtraction using different addressing mode instruction.	3	* ALP to perform addition and subtraction of two given numbers	2	CO3
LLO 4.1 Write an assembly language program for multiplication of two 16 bit unsigned numbers. LLO 4.2 Write an assembly language program for multiplication of two 16 bit signed numbers.	4	ALP for multiplication of two signed and unsigned numbers	2	CO3
LLO 5.1 Write an assembly language program for division of two unsigned numbers. LLO 5.2 Write an assembly language program for division of two signed numbers.	5	ALP to perform division of two unsigned and signed numbers	2	CO3
LLO 6.1 Use DAA and DAS instructions to perform arithmetic operations on BCD numbers. LLO 6.2 Write an ALP to perform arithmetic operations on BCD numbers.	6	ALP to add, subtract, multiply and divide two BCD numbers	2	CO3
LLO 7.1 Implement loop in assembly language program. LLO 7.2 Use string instruction to perform block transfer operation. LLO 7.3 Write an ALP to perform block transfer data without using string instruction. LLO 7.4 Write an ALP to perform block transfer data with using string instruction.	7	* ALP to perform block transfer operation	2	CO4
LLO 8.1 Implement loop in assembly language program to find sum of series. LLO 8.2 Write an assembly language program to find sum of series of n Hexadecimal numbers. LLO 8.3 Write an assembly language program to find sum of series of n BCD numbers.	8	ALP to find sum of series	2	CO4
LLO 9.1 Implement loop in assembly language program to find smallest and largest number from the array of n numbers. LLO 9.2 Use decision making branching instruction to find smallest or largest number. LLO 9.3 Write an assembly language program to find smallest number from the array of n numbers. LLO 9.4 Write an assembly language program to find largest number from the array of n numbers.	9	* ALP to find smallest and largest number from array of numbers	2	CO4
LLO 10.1 Apply iterative method to arrange numbers in array in ascending or descending order. LLO 10.2 Write an assembly language program to arrange numbers in array in ascending order. LLO 10.3 Write an assembly language program to arrange numbers in array in descending order.	10	ALP to arrange numbers in an array in ascending or descending order	2	CO4
LLO 11.1 Write an assembly language program to find length of string. LLO 11.2 Write an assembly language program to concatenate two strings.	11	* ALP to find the length of string and concatenate two strings	2	CO4

MICROPROCESSOR PROGRAMMING**Course Code : 314321**

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 12.1 Write an assembly language program to copy string. LLO 12.2 Write an assembly language program to copy string in reverse order.	12	ALP for string operations such as string reverse and string copy	2	CO4
LLO 13.1 Write an assembly language program to compare two strings without string instruction. LLO 13.2 Write an assembly language program to compare two strings using string instruction.	13	ALP to compare two strings	2	CO4
LLO 14.1 Use div and rotate instructions to check the given number is odd or even. LLO 14.2 Write an assembly language program to count odd and even from the array of n numbers.	14	* ALP to check a given number is odd or even	2	CO4
LLO 15.1 Use rotate instructions to check the given number is positive or negative. LLO 15.2 Write an assembly language program to count positive and negative numbers in given array.	15	ALP to check a given number is positive or negative	2	CO4
LLO 16.1 Use rotate instructions to count '0' and '1' in the given number. LLO 16.2 Write an assembly language program to count number of '0' and '1's in a given number.	16	ALP to count number of '0' and '1's in a given number	2	CO4
LLO 17.1 Use CALL and RET instructions to call procedures using different parameter passing methods.. LLO 17.2 Use assembler directives: PROC and ENDP to write the procedure. LLO 17.3 Write an assembly language program using procedure to perform for addition, subtraction, multiplication and division. LLO 17.4 Write an assembly language program using procedure to solve equation such as $Z = (A+B)*(C+D)$.	17	* ALP to perform arithmetic operations on given numbers using procedure	2	CO5
LLO 18.1 Use assembler directives MACRO and ENDM to write the macros using parameters. LLO 18.2 Write an assembly language program using macro to perform for addition, subtraction, multiplication and division. LLO 18.3 Write an assembly language program using macro to solve equation such as $Z = (A+B)*(C+D)$.	18	ALP to perform arithmetic operations on given numbers using macro	2	CO5

Note : Out of above suggestive LLOs -

- '*' Marked Practicals (LLOs) Are mandatory.
- Minimum 80% of above list of lab experiment are to be performed.
- Judicial mix of LLOs are to be performed to achieve desired outcomes.

VII. SUGGESTED MICRO PROJECT / ASSIGNMENT/ ACTIVITIES FOR SPECIFIC LEARNING / SKILLS DEVELOPMENT (SELF LEARNING)**Micro project**

- The micro project has to be laboratory-based developed in assembly language as suggested by teacher. Each microproject should encompass of two or more CO's which are in fact, an integration of laboratory experiments and

LLO's. Some of the suggested microprojects are given below.

a. Conversion of number system-(Any one):

1. Convert hexadecimal number to equivalent BCD.
2. Convert BCD number to equivalent hexadecimal number

b. Array-(Any one):

1. Separate odd and even number from given array, store them in separate array and find the sum.
2. Separate odd and even number from given array, store them in separate array and find the smallest and largest among them.
3. Separate odd and even number from given array, store them in separate array and sort numbers in ascending and descending order.

c. Basic mathematical functions-(Any one):

1. Generate fibonacci series.
2. Calculate a factorial of given number.

d. String manipulation-(Any one):

1. Convert given lower case string to upper case string and vice-versa.
2. Check the given string for palindrome.
3. Search given character and its position in a string; i.e. find how many times character is present in a string and its position in a string.

Assignment

- Prepare a comparative survey report of 8086 microprocessor with i3, i5, i7, i9 or AMD Ryzen processor.

Note :

- Above is just a suggestive list of microprojects and assignments; faculty must prepare their own bank of microprojects, assignments, and activities in a similar way.
- The faculty must allocate judicious mix of tasks, considering the weaknesses and / strengths of the student in acquiring the desired skills.
- If a microproject is assigned, it is expected to be completed as a group activity.
- SLA marks shall be awarded as per the continuous assessment record.
- For courses with no SLA component the list of suggestive microprojects / assignments/ activities are optional, faculty may encourage students to perform these tasks for enhanced learning experiences.
- If the course does not have associated SLA component, above suggestive listings is applicable to Tutorials and maybe considered for FA-PR evaluations.

VIII. LABORATORY EQUIPMENT / INSTRUMENTS / TOOLS / SOFTWARE REQUIRED

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
1	Hardware: Personal computer, (Processor i3 onwards preferable), RAM minimum 2GB Operating system: Windows-7 onwards	All
2	Software: a) Assembler: Borland Turbo (TASM) / Microsoft Assembler (MASM) b) Linker: Borland Turbo (TLINK) / Microsoft (LINK) c) Debugger: Borland Turbo (TD) / Microsoft debugger (CS or Debug) d) Editor: DOS-Edit / Notepad	All

IX. SUGGESTED WEIGHTAGE TO LEARNING EFFORTS & ASSESSMENT PURPOSE (Specification

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Sr.No	Unit	Unit Title	Aligned COs	Learning Hours	R-Level	U-Level	A-Level	Total Marks
1	I	8086-16 Bit Microprocessor	CO1	6	2	6	6	14
2	II	The Art of Assembly Language Programming	CO2	6	2	2	4	8
3	III	Instruction Set of 8086 Microprocessor	CO3	12	2	8	8	18
4	IV	Assembly Language Programming	CO4	15	0	4	16	20
5	V	Procedure and Macro	CO5	6	2	4	4	10
Grand Total				45	8	24	38	70

X. ASSESSMENT METHODOLOGIES/TOOLS**Formative assessment (Assessment for Learning)**

- Continuous assessment based on process and product related performance indicators
- Each practical will be assessed considering 60% weightage to process 40% weightage to product.

Summative Assessment (Assessment of Learning)

- End semester examination, Lab performance, Viva-voce

XI. SUGGESTED COS - POS MATRIX FORM

Course Outcomes (COs)	Programme Outcomes (POs)							Programme Specific Outcomes* (PSOs)		
	PO-1 Basic and Discipline Specific Knowledge	PO-2 Problem Analysis	PO-3 Design/ Development of Solutions	PO-4 Engineering Tools	PO-5 Engineering Practices for Society, Sustainability and Environment	PO-6 Project Management	PO-7 Life Long Learning	PSO-1	PSO-2	PSO-3
CO1	2	-	-	-	-	1	1			
CO2	2	1	1	2	-	1	1			
CO3	3	2	2	2	-	1	1			
CO4	3	3	3	2	-	1	1			
CO5	3	3	3	2	-	1	1			
Legends :- High:03, Medium:02,Low:01, No Mapping: - *PSOs are to be formulated at institute level										

XII. SUGGESTED LEARNING MATERIALS / BOOKS

Sr.No	Author	Title	Publisher with ISBN Number
1	Douglas V. Hall	Microprocessor and Interfacing (Programming and Hardware)	McGraw Hill Education, New Delhi ISBN-13: 978-0070257429

MICROPROCESSOR PROGRAMMING**Course Code : 314321**

Sr.No	Author	Title	Publisher with ISBN Number
2	Walter A. Triebel, Avtar Singh	The 8088 and 8086 Microprocessors: Programming, Interfacing, Software, Hardware, and Applications	Pearson Publications, New Delhi ISBN-13: 978-0131228047
3	Sunil Mathur	Microprocessor 8086: Architecture, Programming and Interfacing	PHI, New Delhi ISBN-13: 978-8120340879
4	K. R. Venugopal and Raj Kumar	Microprocessor X86 Programming	BPB Publications, Delhi ISBN-13: 978-8170294580

XIII . LEARNING WEBSITES & PORTALS

Sr.No	Link / Portal	Description
1	https://www.tutorialspoint.com/microprocessor/microprocessor_8086_overview.htm	Architecture of 8086
2	https://www.geeksforgeeks.org/architecture-of-8086/	Architecture of 8086
3	https://www.javatpoint.com/8086-microprocessor	Pin description and Architecture of 8086
4	https://electronicsdesk.com/assembler-directives.html	Assembler directives
5	https://www.geeksforgeeks.org/addressing-modes-8086-microprocessor/	Addressing modes of 8086
6	https://www.tutorialspoint.com/microprocessor/microprocessor_8086_addressing_modes.htm	Addressing modes of 8086
7	https://www.tutorialspoint.com/microprocessor/microprocessor_8086_instruction_sets.htm	Instruction set of 8086
8	https://www.javatpoint.com/instruction-set-of-8086	Instruction set of 8086
9	https://nptel.ac.in/courses/108103157	NPTEL Course on Microprocessors and Interfacing

Note :

- Teachers are requested to check the creative common license status/financial implications of the suggested online educational resources before use by the students